Project
Big Data Integration and Processing

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Requirements

Direction 1: Building and install a big data architecture with data integration and processing features

- Dashboard with charts that help understand an overview of the data analysis
- Data analysis: exploring the knowledge/information hidden under the data, helping to solve practical problems.

Collecting data sources that satisfy big data characteristics (minimum 3V volume-velocity-variety), store data

- **Direction 2:**
  - Research in-depth a recent big data integration and processing technology
    - Apache Iceberg, Druid, Deltalake, etc
  - Research in-depth a big data integration and processing papers

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Some directions

- Trend analysis
- Detect unusual/typical events
- Forecast price fluctuations
- Detect fakes and scams
- Evaluate KOLs on social networks
- Analyze sentiments
- Evaluate/forecast potential products
- Real-time big data (IoT)
- Semantic big data integration

Project description

- Title:
  - E.g: apartment rental price analysis in Vietnam
- Data sources
  - E.g: Crawl from internet (list url)
- Requirements/Functions
  - Collecting data: describe data to be collected including key features such as apartment area, location, number of rooms, price…, identify problems/challenges (missing data, irregular data structure, heterogeneity…)
  - Storage requirement: expected volume of data for storage ….
  - Processing/analysis: list the various types of analysis and visualization requirement. E.g calculating the average price over a period, predicting the rental price …. (statistical calculations should be included, and the use of machine learning/AI models is encouraged to discover hidden information within the data and assist in solving the practical problems identified)

Project Timeline

See class schedule